

Unit2

Central Tendencies in Statistics

- **Central Tendencies in Statistics** are the numerical values that are used to represent mid-value or central value a large collection of numerical data.
- These obtained numerical values are called **central** or **average values** in Statistics.

Measures of central tendency

- Mean
 - The average—sum of all values divided by the number of values.
 - Weighted mean: gives different importance (weights) to values.
 - Trimmed mean: ignores extreme values to reduce the effect of outliers

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \mu = \frac{\sum x}{N}$$
$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

2. Median

The middle value when data is ordered.

- Odd number of values → single middle value.
- Even number of values → average of the two middle values.

3. Mode

The most frequently occurring value

- Unimodal: one mode.
- Bimodal: two modes.
- Trimodal: three modes.

Measuring the Dispersion of Data

Quartiles:

- Q1: 25th percentile
- Q3: 75th percentile

Inter-Quartile Range (IQR):

- Formula: $IQR = Q3 - Q1$
- Represents the middle 50% of the data.

Five-Number Summary:

- Minimum, Q1, Median, Q3, Maximum

Boxplot:

- A graphical representation.
- The box represents the IQR.
- Whiskers extend to data points within $1.5 \times IQR$.
- Outliers are plotted as individual points.

Outlier:

- Any value **less than $Q1 - 1.5 \times IQR$** or **greater than $Q3 + 1.5 \times IQR$**

Box Plot Analysis

- "Boxplot Analysis" provides a clear explanation of how a boxplot represents the five-number summary and identifies outliers in a dataset.

Five-Number Summary:

- **Minimum** (Lower Extreme)
- **Q1** (Lower Quartile – 25th percentile)
- **Median** (Middle value – 50th percentile)
- **Q3** (Upper Quartile – 75th percentile)
- **Maximum** (Upper Extreme)

